International Journal of Geographical Information Science

ISSN: 1365-8816 (Print) 1362-3087 (Online) Journal homepage: www.tandfonline.com/journals/tgis20

Scene-GCN: a time-series prediction method in complex monitoring environments through spatial–temporal knowledge graph (ST-KG)

Weihao Li, Yufeng He, Lizhi Tao, Jifa Chen, Xing Wang, Yong Ge & Hui Lin

To cite this article: Weihao Li, Yufeng He, Lizhi Tao, Jifa Chen, Xing Wang, Yong Ge & Hui Lin (2025) Scene-GCN: a time-series prediction method in complex monitoring environments through spatial–temporal knowledge graph (ST-KG), International Journal of Geographical Information Science, 39:10, 2211-2235, DOI: 10.1080/13658816.2025.2479183

To link to this article: https://doi.org/10.1080/13658816.2025.2479183

Published online: 19 Mar 2025.

INTERNATIONAL JOURNAL OF GEOGRAPHICAL INFORMATION SCIENCE
2025, VOL. 39, NO. 10, 2211–2235
https://doi.org/10.1080/13658816.2025.2479183

Taylor & Francis
Taylor & Francis Group

RESEARCH ARTICLE

Scene-GCN: a time-series prediction method in complex monitoring environments through spatial–temporal knowledge graph (ST-KG)

Weihao Li[a,b], Yufeng He[a,b], Lizhi Tao[a,b], Jifa Chen[a,b], Xing Wang[c], Yong Ge[a,b] and Hui Lin[b]

[a]Key Laboratory of Poyang Lake Wetland and Watershed Research, Ministry of Education/School of Geography and Environment, Jiangxi Normal University, Nanchang, China; [b]Early Warning and Assessment of Jiangxi Province, Key Laboratory of Natural Disaster Monitoring, Nanchang, China; [c]School of Atmospheric Sciences, Nanjing University, Nanjing, China

ABSTRACT

Current forecasting methods for complex geographic scenes often fail to adequately account for both intrinsic factors and external environmental variables, which limits their predictive accuracy. This study proposes Scene-GCN, a Graph Convolutional Network (GCN)-based model that utilizes a spatio-temporal knowledge graph (ST-KG) to integrate static and dynamic environmental factors with the scene's spatio-temporal characteristics to predict trends. First, variations at each monitoring target point are abstracted as geo-processes, and instantaneous monitoring values are represented as geographic states, forming a coupled "process-state" ST-KG model tailored for monitoring networks in complex geographic scenes. Environmental knowledge, both dynamic and static, is then embedded within the nodes of the ST-KG, capturing historical and real-time contextual influences through feature computation. Next, a temporal graph convolutional network (T-GCN) cell is developed to quantify the relationships between spatio-temporal features, environmental knowledge, and target predictions. Experimental evaluation on a real-world time-series dataset for monitoring surface deformation in a tailings dam shows that our method outperforms baseline models. Ablation studies further show that the integration of the ST-KG enhances forecasting performance, reducing root mean square error (RMSE) by 49.2% compared to a T-GCN model without ST-KG. Additionally, leveraging a graph neural network framework improves the model's interpretability over traditional time-series forecasting methods.

ARTICLE HISTORY
Received 7 November 2024
Accepted 10 March 2025

KEYWORDS
Geographic scene/VGE; graph convolutional network (GCN); spatial–temporal knowledge representation; time-series forecasting

1. Introduction

The geographic scenes (Geo-scene) Data Model offers a holistic framework for representing the environment (Lü *et al*. 2018, Huang *et al*. 2019, He et al. 2022). In environmental monitoring, a 'scene' refers to a defined spatial area monitored to analyze

CONTACT Yufeng He Yuf.he@jxnu.edu.cn

dynamic processes and detect anomalies. Geographic scene monitoring technologies utilizing GNSS (Li 2011) and the Internet of Things (IoT) (da Cruz *et al*. 2018) are extensively applied in safety-critical fields such as landslide monitoring, subsidence tracking, tailings dam surveillance, and bridge inspection (Metternicht *et al*. 2005, Sun et al. 2012, Qin *et al*. 2018). Accurate trend prediction within these geographic scenes is essential for proactive disaster prevention. It is well known that disasters are influenced not only by the hazard-affected body but also by the surrounding environment. However, few studies incorporate the impact of environmental factors into time-series prediction for hazard-affected bodies (Wang et al. 2024). This gap highlights the urgent need for trend prediction methods in geographic scene monitoring that account for environmental influences.

Currently, various prediction methods that consider temporal dependencies, such as Gradient Boosting Decision Trees (GBDT) (Gao *et al*. 2022), Bayesian inference (Shen *et al*. 2022), and Support Vector Machine (SVM) variants (Ji *et al*. 2022), have been applied to assess risk trends within geographic scenes based on observational data (Gama *et al*. 2020). These methods, although considering the dynamic changes captured by monitoring points within their respective areas in geographic scenes, fail to comprehensively account for the entire geographic scene. Specifically, they overlook the spatial characteristics inherent in the monitoring network formed by monitoring points within complex geographic scenes, consequently omitting the indispensable spatial dependencies within these scenes. To better capture spatial features, Yang *et al*. (2023) utilized convolutional neural networks (CNNs) to model the spatial characteristics of monitoring points. CNNs are deep learning model designed to automatically extract local features from data through convolution operations, widely used in image processing and computer vision tasks. However, CNNs are constrained by Euclidean data structures (e.g. regular grids and images) (Defferrard *et al*. 2016), making it challenging to accurately represent the complex network relationships among monitoring points within geographic scenes. Consequently, these methods are limited in characterizing the spatial dependencies inherent to such networks.

In contrast, graph neural networks (GNNs) allow real-world problems to be abstracted into graph-structured data, capturing dependencies by passing messages between graph nodes (Zhou *et al*. 2020). GNNs are deep learning models designed to effectively process and analyze graph-structured data by capturing relationships and dependencies between nodes through message passing mechanisms. Lin *et al*. applied the graph attention network (GAT) (Lin *et al*. 2023) to model the spatial dependencies among monitoring points within complex geographic scenes, yielding promising prediction results. This approach integrates CNNs into the GNN framework, enabling efficient processing of data from non-Euclidean domains (Yao *et al*. 2019). These advances underscore the importance of spatio-temporal methods in complex geographic scene monitoring, particularly those that integrate environmental influences for enhanced trend prediction and proactive disaster prevention.

However, accurate trend prediction of monitoring targets in complex geographic scenes requires not only the integration of the spatial–temporal characteristics associated with target evolution but also a comprehensive incorporation of the various factors influencing observations from the monitoring network. The trend of these observations is governed not only by environmental changes (i.e. external dynamic factors) but also by the intrinsic characteristics of the scene itself (i.e. static factors). For example, in a tailings dam monitoring system, static factors such as soil composition, rock structure, and slope orientation remain constant over time. In contrast, dynamic factors like precipitation fluctuate, exerting substantial influence over extended periods. Rainfall can lead to the gradual accumulation of moisture on the dam surface, resulting in osmotic erosion that progressively undermines the dam's structural stability (Silva Rotta *et al*. 2020). These combined influences introduce considerable uncertainty in forecasting for monitoring targets such as tailings dams, presenting a significant challenge for precise trend prediction in complex geographic scenes (Jiang *et al*. 2024). To structure both dynamic and static scene knowledge, a spatio-temporal knowledge graph that integrates entities and relationships offers an efficient approach for organizing scene knowledge (He *et al*. 2025), enabling advanced knowledge mining and prediction.

To comprehensively account for monitoring networks in complex geographic scenes, this paper proposes a spatial–temporal forecasting method named Scene-GCN. This approach integrates various factors influencing each monitoring point as dynamic and static environmental knowledge, addressing the issue in complex geographic scene modeling where only monitoring targets are considered, yet the inherent connections between targets and scenes are overlooked. Additionally, it further integrates the temporal dependence of historical time-series monitoring data and the spatial dependence of the monitoring location. The main contributions of our study are as follows:

1. To construct a spatial–temporal knowledge graph (ST-KG), dynamic and static environmental knowledge in complex geographic scenes is embedded as attributes into historical time-series monitoring data. Subsequently, a weighted undirected connected(adjacency) graph is established to anchor the spatial structure of the monitoring network within the scene, facilitating the learning of network topology within the graph model. By fully considering the lagged effects of environmental factors in the surrounding area, the attribute embedding in ST-KG enhances the model's performance by synthesizing data from neighboring points and various influencing elements, thus increasing the interpretability of the forecasting method and improving its predictive accuracy.
2. The T-GCN cell (Zhao *et al*. 2020) is used to extract feature information from the ST-KG and realize spatial–temporal forecasting. The spatial dependencies between nodes in the ST-KG are captured by the GCN, while the gated recurrent unit (GRU) is utilized to analyze the temporal dependencies in the time-series data and capture dynamic changes within the ST-KG.

The remainder of this paper is structured as follows: Section 2 provides a review of relevant forecasting studies on monitoring networks within complex geographic scenes. Section 3 presents the methodological framework in detail. In Section 4, experiments using real-world datasets are conducted to evaluate the proposed model's performance compared to baseline models. Finally, Section 5 summarizes the findings and discusses prospects for future research.

2. Related work

With the rapid advancement of technology, the application of advanced methods to improve the efficiency and accuracy of prediction and early warning systems in complex geographic scenes has become a prominent research focus.

Among previously proposed approaches for monitoring network forecasting in complex geographic scenes, Physical modeling methods primarily involve profiling the engineering structures within complex geographic scenes and modeling and analyzing the physical processes occurring in the surrounding geological environment (Yin *et al*. 2011). Representative methods include finite element analysis (FEA) (Krahn 2003), computational fluid dynamics (CFD) (Gao and Fourie 2015), and the inverse-velocity method (Rose and Hungr 2007). All these physical modeling approaches have achieved significant advances in forecasting research within complex geographic scenes. However, the physical properties and environmental influences vary across different complex geographic scenes, making it challenging to develop a universal physical model that can account for the distinct characteristics of each scene. This limitation arises from the numerous implicit physical relationships among variables that directly or indirectly affect the physical system, many of which are likely unknown (Elsheikh *et al*. 2019).

In contrast, data-driven approaches rely on large volumes of monitoring network data within complex geographic scenes to perform forecasting tasks, eliminating the need for detailed physical analysis of the scenes. Although these approaches do not account for the physical properties of the environment or provide explanations for the underlying processes that drive changes within these scenes, they have become increasingly popular due to their adaptability and high predictive accuracy. With the increasing body of research related to prediction and early warning in complex geographic scenes, numerous high-accuracy prediction methods have emerged, which can be broadly classified into two categories: parametric and nonparametric models (Vlahogianni *et al*. 2004, Van Lint and Van Hinsbergen 2012). Parametric models are a class of statistical models that assume a fixed structure for the data distribution. These models achieve the task of monitoring target forecasting by completely describing the distribution of the data through a set of parameters. Time series models and linear regression models (Diskin 1970, Mendoza *et al*. 2021) are common examples of parametric methods. Nie *et al*. and Bui *et al*. applied these methods to geotechnical engineering-related forecasting tasks (Bui et al. 2019, Nie *et al*. 2022). Traditional parameterized model algorithms are straightforward to interpret; however, these methods involve assumptions about data distribution that cannot sufficiently capture the nonlinearity and uncertainty in forecasting within complex geographic scenes (Li et al. 2019). In contrast, nonparametric models address this issue effectively and can capture complex data patterns when sufficient historical data are available. Examples of nonparametric models include support vector regression, k-nearest neighbors, and neural networks (Elshorbagy *et al*. 2010, Salazar *et al*. 2015). Deep learning methods (Silver *et al*. 2016, Moravčík *et al*. 2017) possess superior fitting capabilities for modeling arbitrarily complex nonlinear relationships and demonstrate strong compatibility when integrated with traditional physical modeling methods (Kochkov *et al*. 2024).

Consequently, these methods have been widely applied in prediction and early warning within complex geographic scenes (Yang *et al*. 2019).

The models can be categorized into two groups based on whether they consider spatial correlation. Some forecasting methods focus solely on temporal dependence, and several studies have achieved improved forecasting accuracy by training models such as long short-term memory (LSTM) (Zhu *et al*. 2022) and GRU (Zhang *et al*. 2022). However, these models consider only temporal dependencies and overlook the spatial characteristics in complex geographic scenes, resulting in predictions that do not account for the spatial constraints inherent in monitoring networks. Fully leveraging the spatial–temporal dependencies within monitoring networks in complex geographic scenes is essential for enhancing related forecasting tasks. To better capture spatial features, some researchers have employed convolutional neural networks (CNNs) to establish spatial correlations within monitoring networks (Yang *et al*. 2020, Hua *et al*. 2023).

The methods described above incorporate CNNs in the process of establishing spatial dependence and have shown notable progress in forecasting tasks. However, the CNN approach primarily relies on Euclidean spatial features, such as images and regular grids, making it ineffective at capturing or applying non-Euclidean spatial features, including topological relationships and interrelationships. This limitation hinders its ability to accurately reflect the true spatial dependencies within monitoring networks in complex geographic scenes. With the advent of the graph convolutional network (GCN) approach (Kipf and Welling 2016), some studies have leveraged this network to enhance the capture of spatial features within graph-structured data (Zhao *et al*. 2020, Ma et al. 2021, Ruan *et al*. 2023).

In complex geographic scenes, the development of GCN models for monitoring network forecasting has been constrained due to their insufficient integration of the spatial–temporal characteristics associated with the evolution of monitored targets. Moreover, these models rarely incorporate the dynamic and static environmental knowledge of the scene, limiting their ability to interpret changes in monitoring targets. The T-GCN model (Zhao *et al*. 2020) has been proven effective in handling the spatial–temporal features of urban traffic networks but has yet to be practically applied in the monitoring and forecasting of complex geographic scenes. To address the lack of dynamic and static environmental knowledge within the scene, He *et al*. (2025) proposed a seven-element data model encompassing time, space, event, process, state, and human/object to depict the dynamic spatial–temporal evolution of a geographic scene and developed an ST-KG to represent environmental influences on monitored targets. Therefore, in our Scene-GCN framework, we incorporate the ST-KG construction theory from the aforementioned work and further integrate the T-GCN model to achieve predictions. These predictions not only improve accuracy but also enhance interpretability.

3. Methodology

3.1 Framework

The Scene-GCN model integrates the spatio-temporal features of complex geographic scenes with dynamic and static environmental knowledge through the ST-KG

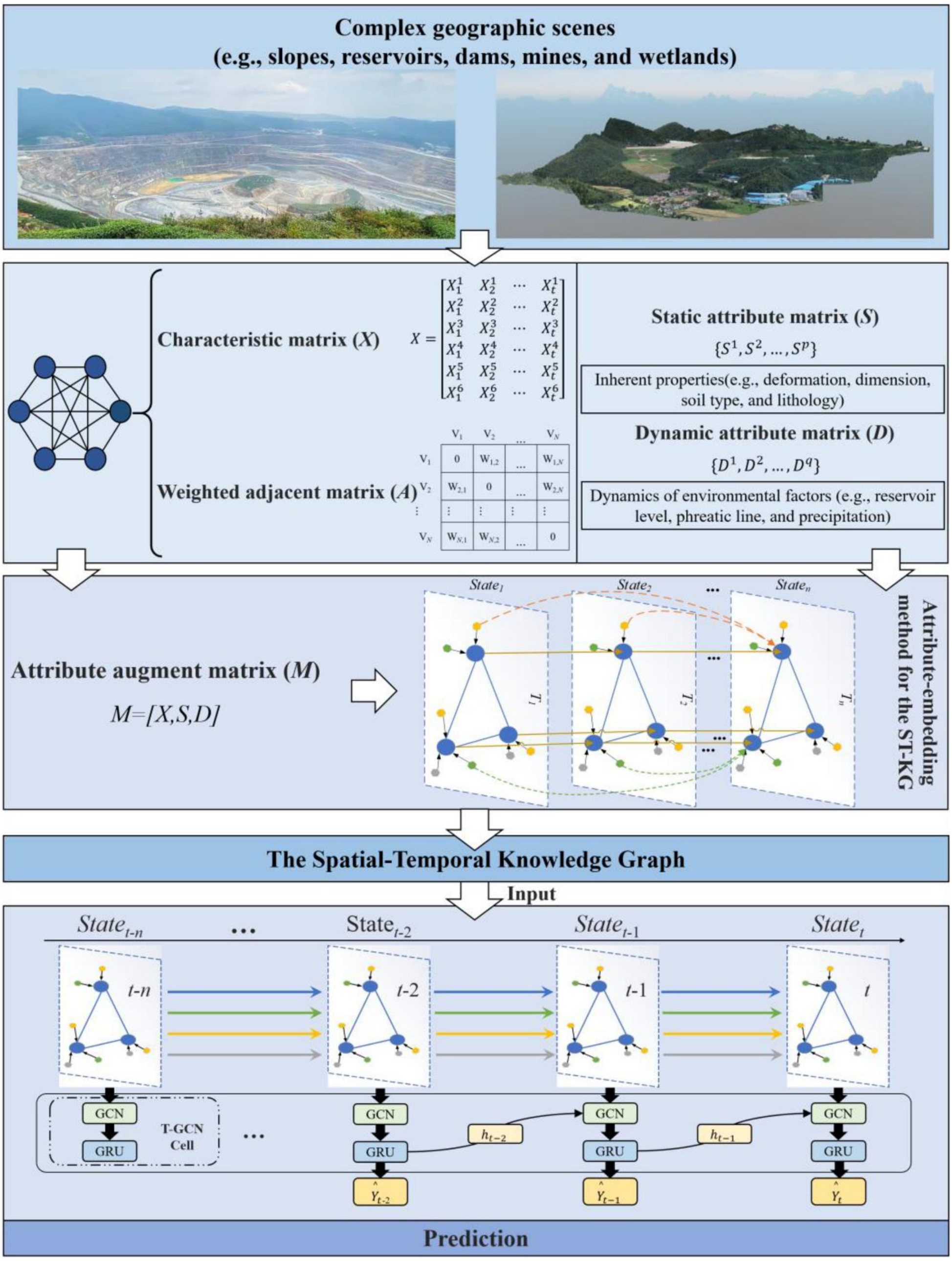

Figure 1. Scene-GCN spatio-temporal forecasting modeling framework.

approach and employs T-GCN units to perform spatio-temporal predictions. As shown in Figure 1, taking deformation monitor scene as example, First, the monitoring targets in complex geographic scenes (e.g. surface deformation) are represented as the feature matrix X, and use a weighted adjacency matrix A to model the spatial dependencies among the monitoring targets. Second, in complex geographic monitoring scenes, observational data from IoT-based sensor networks are subject to influence from the dynamic and static environmental factors. On the one hand, dynamic changes in the environment around monitors (e.g. the dry beach length, reservoir level, phreatic line, and precipitation in the monitoring environment) influence the stability of the monitoring target (Yin *et al*. 2011, Dong *et al*. 2020, Hancock 2021, Ruan *et al*. 2023). On the other hand, the static environmental factors are related to the intrinsic properties of the monitoring target (e.g., the deformation dimensions, soil type, and lithology in the monitoring environment), also affect the stability of the monitoring target (Sun et al. 2012, Naeini and Akhtarpour 2018, Du *et al*. 2022). Therefore, the dynamic and static environmental factors are represented using a dynamic attribute matrix D and a static attribute matrix S, respectively. Third, the feature matrix X is combined with the two attribute matrices S and D to form the attribute-enhanced matrix M. Subsequently, the relationships between attributes and features in matrix M are established through the attribute embedding method of the ST-KG, thereby constructing the ST-KG. Fourth, for each timestamp of the ST-KG, the proposed method extracts the feature values from the feature matrix X, the spatial dependencies corresponding to these feature values in the weighted adjacency matrix A, and the attribute values associated with these feature values in the attribute matrices S and D. After the extraction, the spatial dependencies of the feature values within each timestamp and the temporal dependencies among feature values, attribute values, and the interactions between feature values and attribute values across different timestamps are established. Finally, spatio-temporal predictions are performed using the T-GCN units.

Therefore, based on the above definitions, examples, and the framework illustrated in Figure 1, the time-series forecasting problem can be defined as learning the function f to predict the time-series data at a future time, given the network topology A of the monitors, the feature matrix X, the attribute matrices S and D, the attribute-enhanced matrix M, and the spatial–temporal knowledge graph (ST-KG):

$$y = f(A, M = (X, S, D), ST - KG) \tag{1}$$

3.2 Spatial–temporal knowledge graph

In complex geographic scenes, IoT-based sensor networks encompass various monitors, including deformation monitors, water level monitors, and precipitation monitors. To model the interactions between monitoring targets and both dynamic and static environmental factors, the ST-KG construction method (as shown in Figure 2) is developed as follows:

Assuming the monitoring target is deformation within the scene, with environmental factors such as precipitation and water level represented as attributes. (1) Each monitoring target is conceptualized as an entity, with the temporal evolution of each entity abstracted as a process, and deformation values at different time points represented as states. Spatial dependencies are established by quantifying the spatial structure between entity states, allowing not only for consideration of temporal dependencies within individual processes but also for the extension of temporal dependencies across multiple processes. (2) Attributes of the monitoring target are classified into dynamic and static types, as the evolution of these attributes may

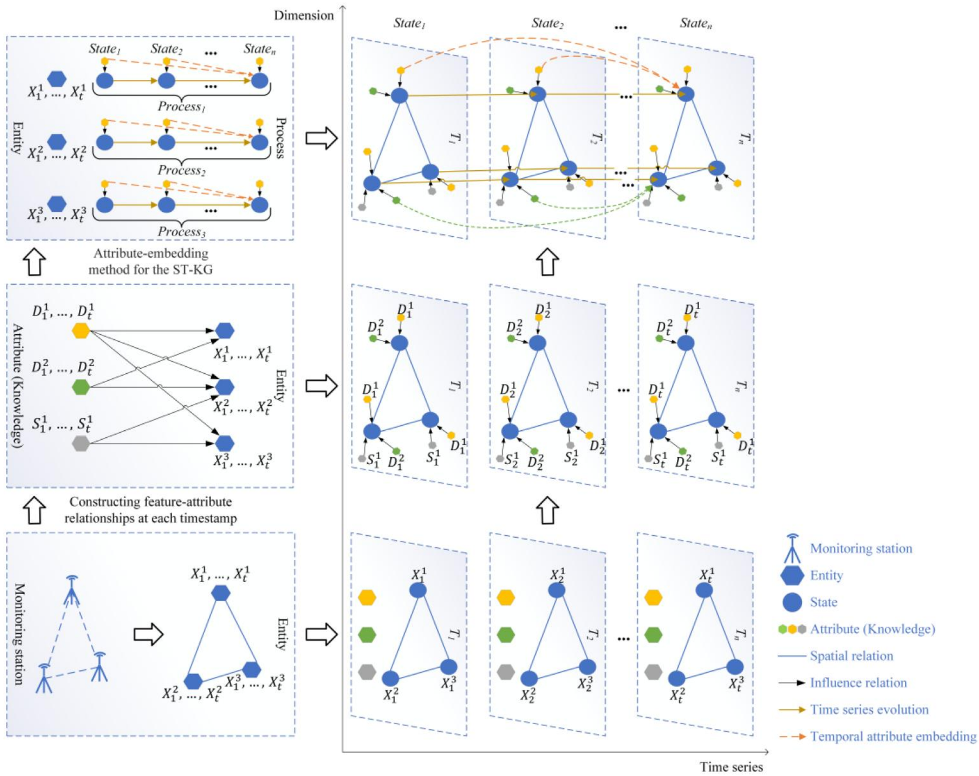

Figure 2. Schematic diagram of the ST-KG construction method.

amplify or attenuate the deformation values of the monitoring target. Consequently, embedding attributes into entities significantly enhances the graph's capacity for knowledge representation.

(1) Abstraction and representation of deformation-related spatial–temporal knowledge

To express the spatial dependence among the studied monitor, a weighted, undirected, connected graph $G = (V, E, W)$ is defined in this paper, as shown in Figure 3. The monitors are abstracted as graph nodes, i.e., $V = v_1, v_2, \ldots, v_N$, where N is the number of monitors. Each pair containing any two monitors has one undirected edge E. The total number of undirected edges E is given by $E = N \times (N-1)/2$. $W \in R^{N \times N}$ denotes the adjacency matrix based on the spatial proximity of all monitors. The spatial distance between two monitors relates to their time-series data and represents an important factor affecting the spatial dependence between the points (Ma *et al.* 2021). The Gaussian similarity function based on spatial proximity is used to calculate the weights of the weighted adjacency matrix (Von Luxburg 2007), the formula of which is given below:

$$w(i,j) = exp\left(-\| v_i - v_j \|^2 / 2\sigma^2\right) \tag{1}$$

where $w(i,j)$ is the weight size of the undirected edge e_{ij} between two monitors (v_i, v_j), $\| v_i - v_j \|^2$ is the spatial distance between the two monitors (v_i, v_j), and σ is the standard deviation of the set of distances of each monitoring point.

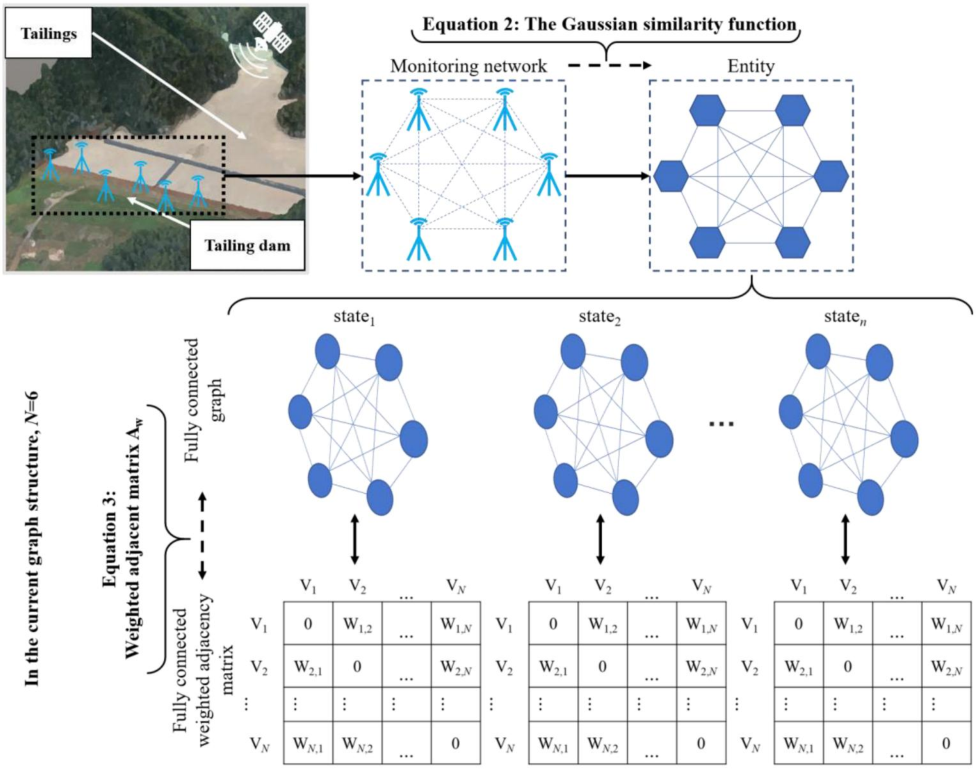

Figure 3. The weighted connected graph of the monitoring network in complex geographic scenes.

The weighted adjacency matrix among the monitors is shown below:

$$A_w = \begin{bmatrix} 0 & w(1,2) & \cdots & w(1,N) \\ w(2,1) & 0 & \cdots & w(2,N) \\ \vdots & \vdots & \ddots & \vdots \\ w(N,1) & w(N,2) & \cdots & 0 \end{bmatrix} \tag{2}$$

To express the temporal dependence of monitors, a feature matrix $X \in R^{N\times L}$ including the time-series information of all monitors is constructed. Here, N denotes the number of monitors, and L is the total length of the time series. $X \in R^{N\times t}$ represents the observed values of all monitors at time t. If n historical time-series data points are given, this is expressed as:

$$[X_{t-n}, \ldots, X_{t-1}, X_t] \tag{3}$$

Accordingly, the forecast of the future T time series lengths are:

$$[X_{t+1}, \ldots, X_{t+T-1}, X_{t+T}] \tag{4}$$

(2) Attribute-embedding method for the ST-KG

The intrinsic properties of the monitoring target are taken as the static attributes (S), and dynamic changes in the environmental factors are taken as the dynamic attributes

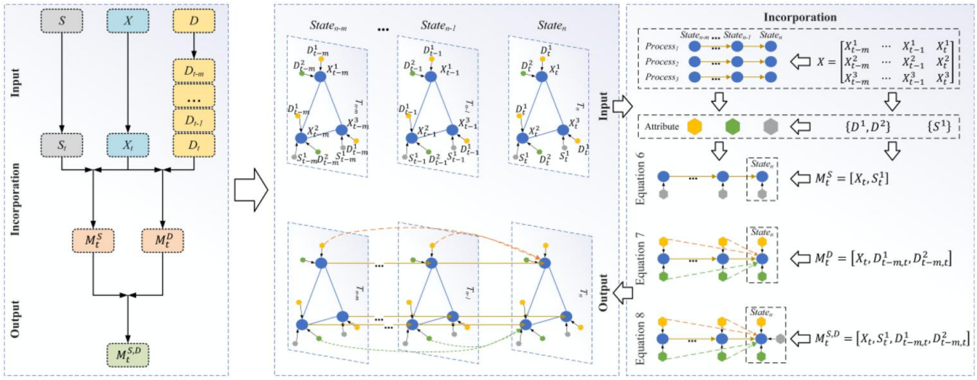

Figure 4. Construction of the attribute matrix.

(D). Considering the time-series state entity X, the process of embedding attributes S, D into state X is shown in Figure 4.

Embedding the static attributes: Let $S \in R^{N\times p}$ be the set of p distinct static attributes $\{S^1, S^2, \ldots, S^p\}$. These static attributes do not change over time, and the constant values of S are used to generate a matrix of the static attributes M_t^S for each time point. Specifically, the matrix of static attributes at timepoint t is defined below:

$$M_t^S = [X_t, S], M_t^S \in R^{N\times(p+1)} \tag{5}$$

Embedding the dynamic attributes: Unlike S, let $D \in R^{N\times(q*t)}$ be the set of q distinct dynamic attributes $\{D^1, D^2, \ldots, D^q\}$. Considering the time lag in the observed values relative to changes in external environmental factors, when constructing the dynamic attribute matrix M_t^D, the size of the selection window is extended to $m+1$ to replace the attribute corresponding to the original time t, i.e., the dynamic attribute matrix $D_{t-m,t}^q = [D_{t-m}^q, \ldots, D_{t-1}^q, D_t^q]$ is chosen as the dynamic attribute submatrix D^q. Specifically, the dynamic attribute matrix at time t is defined below:

$$M_t^D = \left[X_t, D_{t-m,t}^1, D_{t-m,t}^2, \ldots, D_{t-m,t}^q\right], M_t^D \in R^{N\times[q*(m+1)]} \tag{6}$$

Finally, the combined representation of the feature matrix X, the static attributes S, and the dynamic attributes D combined with the attribute matrix $M_t^{S,D}$ at time t can be defined as follows:

$$M_t^{S,D} = \left[X_t, S, D_{t-m,t}^1, D_{t-m,t}^2, \ldots, D_{t-m,t}^q\right], M_t^{S,D} \in R^{N\times[p+1+q*(m+1)]} \tag{7}$$

where q denotes the number of dynamic attributes.

3.3 Time-series forecasting method based on the Scene-GCN

The ST-KG described above establishes a graph data model for the non-Euclidean domain, while the GCN (Wu *et al.* 2021) can effectively capture the spatial dependence in non-Euclidean domain data. When a node's features are updated, this model can comprehensively consider the characteristics of node's neighbors (and those of the node itself). The sigmoid function is used as the activation function for the GCN to

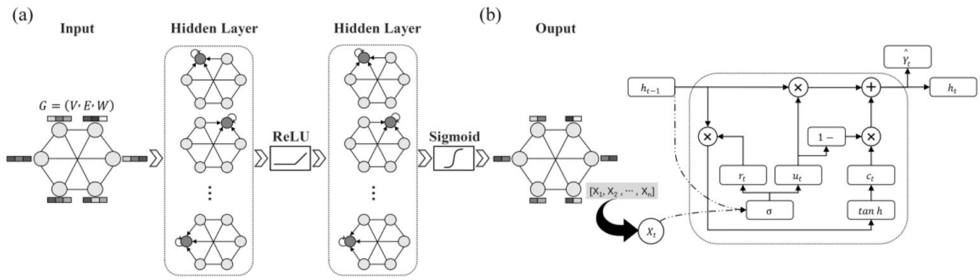

Figure 5. Structure of the two deep learning model. (a) The graph convolutional network structure. (b) The gated recurrent unit structure.

obtain the features with the spatial dependence from all the nodes by updating the feature information in the graph structure (Kipf and Welling 2016, Zhao *et al*. 2020).

As shown in Figure 5(a), to obtain the spatial dependence between nodes, the GCN takes the adjacency matrix A and feature matrix X as inputs, whose layer-by-layer propagation rule is given by:

$$Y^{(l+1)} = \sigma\left(\tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}Y^{(l)}W^{(l)}\right) \tag{8}$$

where $\tilde{A} = A + I_N$ is the self-connected adjacency matrix of the undirected graph G, and I_N is the identity matrix; $\tilde{D} = \sum_j \tilde{A}_{ij}$ is the degree matrix of $\tilde{A}$; $W^{(l)}$ is the layer l trainable weight matrix; $\sigma(\cdot)$ denotes the sigmoid function, a non-linear activation function commonly used in neural network models; $Y^{(l)} \in R^{N\times D}$ is the matrix activated in layer l, $Y^{(l+1)}$ is the output of the convolutional layer, and $Y^{(0)} = X$.

In this study, the two-layer GCN model is used to capture spatial dependencies, and its expression is as follows:

$$f(X, A) = \sigma\left(\hat{A}\ \ ReLU\left(\hat{A}XW_0\right)W_1\right) \tag{9}$$

where $\hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}$ represents the preprocessing step, $W_0 \in R^{P\times H}$ denotes the weight matrix from the input layer to the hidden layer, P is the length of the feature matrix and H is the number of hidden units. $W_1 \in R^{H\times T}$ denotes the weight matrix from the hidden layer to the output layer, T is the prediction length. $ReLU()$ represents the Rectified Linear Unit, a widely used activation function in modern deep neural networks.

The GRU has a gating unit to regulate information flow inside the unit. There is no separate storage unit; thus, the GRU is simpler in structure and more efficient in its training time and update optimization compared to the LSTM approach (Chung *et al*. 2014, Cho *et al*. 2014).

As shown in Figure 5(b), the input feature matrix in the node at the current time is X_t; the hidden state at the previous moment is h_{t-1}; the reset gate r_t is used to control the extent to which historical information is ignored; the update gate u_t is used to control the extent to which historical information is combined with the current state of inputs; c_t is the candidate memory cell, that is, the content of the information storage at time t; h_t is the hidden state of the output at time t; $\hat{Y}_t$ is the forecast of observed values at time t. The GRU uses the current monitoring information at time t and the hidden state information at $t-1$ as inputs; thus, when capturing the monitoring information at the current moment, it can also consider trends in historical information to synthesize the output results.

If the current time is t, M_t is the corresponding attribute matrix extracted at that time point; this is then imported into the GCN layer for feature updating and time-series changes with spatial dependence are produced as output features. These feature sequences are then used as inputs to the GRU to construct the temporal dependence of the sequences in the gating unit, which ultimately realizes the time-series forecasting task. The derivation process is as follows:

$$r_t = \sigma(W_r \cdot [gc(A, M_t), h_{t-1}] + b_r) \tag{10}$$

$$u_t = \sigma(W_u \cdot [gc(A, M_t), h_{t-1}] + b_u) \tag{11}$$

$$c_t = \tan h[gc(A, M_t), (r_t * h_{t-1})] + b_c) \tag{12}$$

$$h_t = u_t * h_{t-1} + (1 - u_t) * c_t \tag{13}$$

where $gc(\cdot)$ is the graph convolution process, $\sigma(\cdot)$ and $\tan h(\cdot)$ are the activation functions, and the parameters W and b are the learnable weights and biases, respectively.

4. Experiments and analysis

4.1 Overview of the study area

The model proposed in this paper was validated through a case study of the monitoring network in the complex geographic scene of the Dexing copper mine tailings dam in China. The Dexing copper mine has a total area of 37 km^2 and is situated in the middle and lower reaches of the Yangtze River; this mine belongs to Dexing Township, Shangrao City, northeastern Jiangxi Province, which is located in the Huayu Mountains (Zhang *et al*. 2021, Du *et al*. 2022). The No.4 tailings pond is situated at the boundary of the open pit area of the Dexing Copper Mine and has the largest volume, surface area, and dam volume. The structure, surrounding environment, and monitoring network of this complex geographic scene are shown in Figure 6.

4.2 Dataset description and processing

The dataset used in this study consists of monitoring network data from August 25, 2022, to November 6, 2023, for the Dexing Copper Mine No. 4 tailings pond. The deformation is described by X, Y, Z 3-D surface deformation data, while the data types

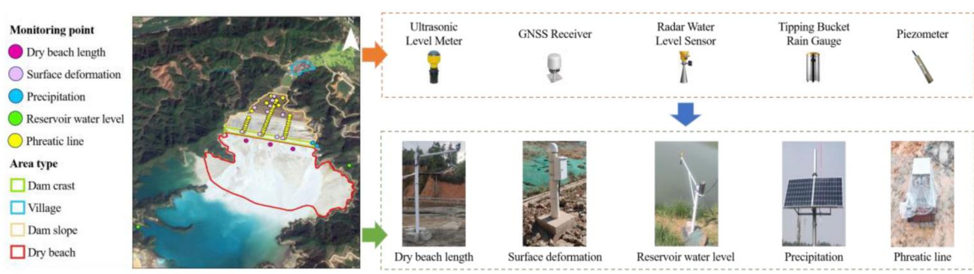

Figure 6. Schematic diagram illustrating the monitoring network distribution at the No. 4 tailings pond.

of the external environmental factors are the dry beach length, reservoir water level, phreatic line, and precipitation. Using those features, the following workflow was applied:

1. Construct the feature matrix: to comprehensively consider the interaction among the long time-series deformation features at each deformation monitoring point, deformation data in the X, Y, and Z directions from 10 surface deformation monitoring points are used, resulting in a total of 30 time series. Each time series contains 21,027 time points, with an interval of 30 minutes between two consecutive time points. The size of the corresponding matrix is 30*21027.
2. Construct the adjacency matrix: because all the monitors influence among each other, fully connected graph is used in this experiment. the distance weight matrix based on the Gaussian similarity function is constructed for the studied deformation monitors. The size of this matrix is 30*30.
3. Construct the static attribute matrix: the X, Y, Z dimensions are used as static attributes. The size of this matrix is 30*1.
4. Construct the dynamic attribute matrix: three dry beach length monitors, two reservoir water level monitors, 45 phreatic line monitors, and two precipitation monitors were used, and the time-series interval was synchronized with the deformation data. In order to synthesize the influence of the external factors on the study area, the data were processed to obtain the average dry beach length, average reservoir water level, average phreatic line, and maximum precipitation at each time point. The size of the corresponding matrix containing these data is 4*30*21027.

The constructed feature matrix, static attribute matrix, and dynamic attribute matrix were then linearly transformed from the original data using min-max normalization such that the normalized results were mapped to the range $[0, 1]$ using the following transformation equation:

$$X = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{14}$$

where $x_{\max}$ is the maximum value in the matrix, $x_{\min}$ is the minimum value in the matrix, x is the original data point in the matrix, and X is the normalized result.

4.3 Model training and parameterization

(1) Training rules

The dataset division approach is shown in Figure 7. As shown, 80% of the values in the dataset were used to form the training set, while the remaining 20% were used as the test set. The samples from the training set were extracted using a sliding window approach in the forward direction of the time series. Considering the practical requirements of tailings dam failure early warning research, it is necessary to quickly predict short-term changes in the dam body to enable timely emergency measures. Therefore, a short-term prediction approach was adopted to enhance the model's real-time responsiveness. In this approach, six training samples were extracted each time; of

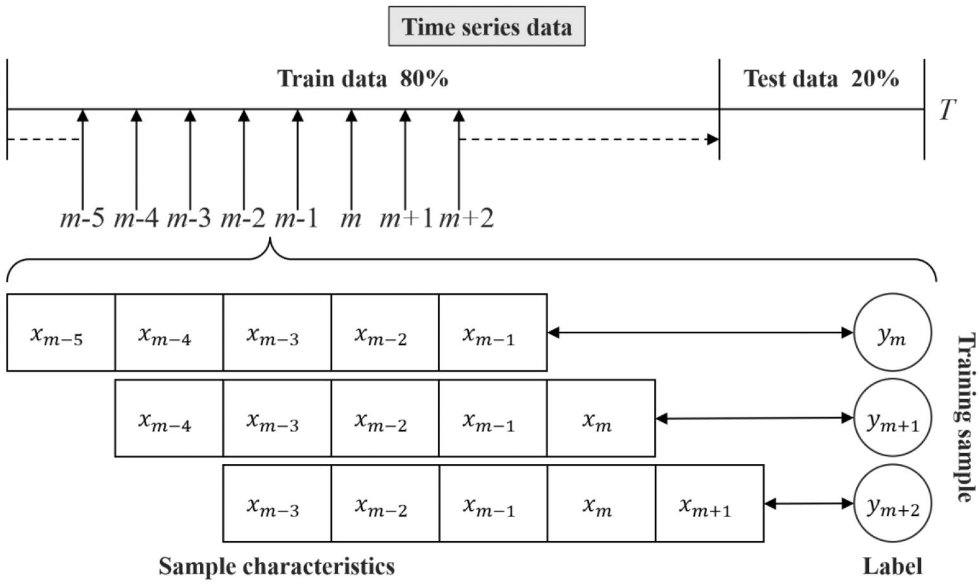

Figure 7. Dataset segmentation.

these, the first five were used as sample inputs and the sixth was used for labeling, with a sliding step size of one. The test set was processed in the same way.

The goal of the tailings dam deformation prediction is to ensure that the predicted results are as close as possible to the true deformation state of the tailings dam. Therefore, the loss function aims to minimize the forecast error of the model, and the loss function used for the training process is defined as:

$$loss = \sum_{t=1}^{n} (Y_t - \hat{Y}_t)^2 / n + \lambda L_{reg} \tag{15}$$

where n is the total number of training samples, Y_t is the ground truth value of the tailings dam deformation, $\hat{Y}_t$ is the predicted value of the tailings dam deformation, L_{reg} is the L2 regularization term that helps to avoid overfitting problems, and λ is the hyperparameter that adjusts the degree of regularization

(2) Evaluation metrics

The overall performance of the model was evaluated based on the following five metrics.

a. Root mean square error (RMSE):

$$RMSE = \left[\frac{1}{n}\sum_{t=1}^{n} (Y_t - \hat{Y}_t)^2\right]^{\frac{1}{2}} \tag{16}$$

The closer the RMSE value is to 0, the smaller the model's forecasting error is; thus, smaller values indicate that the model's forecast is more accurate.

b. Mean absolute error (MAE):

$$MAE = \frac{\sum_{t=1}^{n} |Y_t - \hat{Y}_t|}{n} \tag{17}$$

The MAE metric is used to assess the degree of closeness between the model's predictions and the real dataset: the closer its value is to 0, the better the model is fitted.

c. Accuracy:

$$Accuracy = 1 - \frac{\| Y - \hat{Y} \|_F}{\| Y \|_F} \tag{18}$$

where Y and $\hat{Y}$ represent the sets of Y_t and $\hat{Y}_t$, respectively, and $\| \cdot \|_F$ represents the Frobenius norm. The Frobenius norm normalizes the error ($\| Y - \hat{Y} \|_F$) relative to the overall scale of the actual data ($\| Y \|_F$), ensuring that the error and accuracy are comparable across datasets with different scales or units. The closer the value is to 1, the better the performance of the model.

d. R^2:

$$R^2 = 1 - \frac{\sum_{t=1}^{n} (Y_t - \hat{Y}_t)^2}{\sum_{t=1}^{n} (Y_t - \overline{Y}_t)^2} \tag{19}$$

The R^2 metric is mainly used to measure the model's forecasting ability. Where $\sum_{t=1}^{n} (Y_t - \hat{Y}_t)^2$ represents the sum of squared residuals, which measures the error introduced by the predictions, and $\sum_{t=1}^{n} (Y_t - \overline{Y}_t)^2$ represents the total sum of squares (TSS), which measures the total variance in the actual data. Here, $\overline{Y}_t$ denotes the mean value of the observed data. The closer the R^2 value is to 1, the better the forecasting results.

e. Explainable variability (VAR):

$$VAR = 1 - \frac{var(Y - \hat{Y})}{var(Y)} \tag{20}$$

VAR measures the proportion of the model's predictions that account for actual tailings dam deformation. Where $var(Y - \hat{Y})$ represents the variance of the error between the predicted values and the actual values, and $var(Y)$ represents the overall variance of the actual data. The closer the value of this metric is to 1, the better the model's predictive ability.

(3) Parameter setting

In this paper, the main hyperparameters of the proposed model include the training epoch, batch size, learning rate, and the number of GRU hidden units. In the experiments, a small batch size resulted in larger gradient fluctuations during updates, while an excessively large batch size increased the risk of the model getting stuck in local optima. And a small learning rate could lead to convergence at local optima, whereas a large learning rate might cause the model to miss the global optimum. Taking these factors into account, we manually adjusted the parameters and set the batch size to 64 and the initial learning rate to 0.001. Furthermore, the Adam optimizer (Kingma and Ba 2014) was employed during model training to dynamically and adaptively adjust the learning rate. Notably, the number of hidden units of the model is crucial

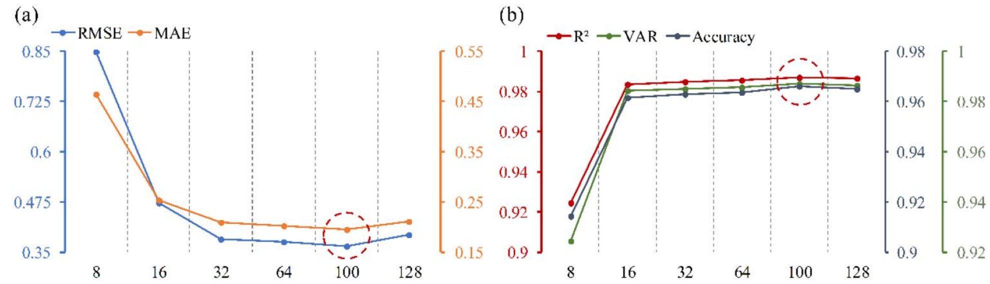

Figure 8. Comparison of the model's predicted performance on the test set with different hidden units. (a) Changes in RMSE and MAE. (b) Changes in accuracy, R^2, and VAR.

to the accuracy of its forecasting; thus, this parameter was determined through multiple experiments.

The number of GRU hidden units was selected from the range [8,16,32,64,100,128] to analyze the variations in forecast accuracy. In Figure 8, the horizontal axis represents the number of hidden units, and the vertical axis represents the variations in different evaluation metrics. Figure 8(a) shows the variations of RMSE and MAE in the test set, where the error is minimized when the number of hidden units is 100. Similarly, Figure 8(b) shows variations in the accuracy, R^2, and VAR metrics on the test set, with the best values of all three evaluation metrics achieved when the number of hidden units was set to 100. It is worth noting that in both Figure 8(a) and (b), as the number of hidden units increases, the prediction precision firstly increases and then decreases. This is primarily because when the number of hidden units exceeds a certain threshold, the complexity and computational difficulty of the model significantly increase, leading to overfitting on the training data. Considering these factors, we set the number of hidden units to 100. And the value of the training epoch was set in the same way as for the GRU hidden unit and finally adjusted to 2500.

4.4. Experimental results and analysis

In this section, the experimental results for 30 time-series datasets of monitoring targets within the monitoring network in complex geographic scenes are analyzed from two perspectives: (1) comparison with other baseline methods using the assessment metrics described above and (2) ablation experiments to validate the utility of the Scene-GCN approach in tailings dam deformation forecasting.

(1) Comparison of experimental results

We compare the performance of the proposed model in this paper to the following baseline methods for time-series forecasting: the autoregressive integrated moving average model (ARIMA) (Percival and Walden 1993), the support vector regression model (SVR) (Smola and Schölkopf 2004), the GRU model (Cho *et al*. 2014), the attention temporal graph convolutional network (A3T-GCN) (Bai *et al*. 2021), and the T-GCN model (Zhao *et al*. 2020). In the baseline model considering spatial dependence, the adjacency matrix construction process is identical to the model proposed in this paper. The forecast results of the various models are shown in Table 1.

Table 1. The prediction results of the Scene-GCN and other baseline methods on the Dexing No. 4 tailings pond dataset.

Evaluation metrics	ARIMA	SVR	GRU	A3T-GCN	T-GCN	Scene-GCN ST-KG (Static + Dynamic)
RMSE	14.741	1.813	0.941	0.682	0.720	**0.366**
MAE	14.730	1.052	0.519	0.344	0.362	**0.196**
Accuracy	*	0.945	0.945	0.946	0.944	**0.965**
R^2	*	0.969	0.969	0.970	0.968	**0.987**
VAR	*	0.970	0.969	0.971	0.968	**0.987**

*indicates that the values are small enough to be negligible, indicating that the model has poor forecasting capability.

Table 2. Ablation Experiments under different experimental settings. (a) Unembedded ST-KG vs. embedded single attribute (b) Unembedded ST-KG vs. model embedded with different attributes (all static attributes/all dynamic attributes/all static + all dynamic attributes).

(a) Unembedded ST-KG vs. embedded single attribute

		Scene-GCN				
			ST-KG (Dynamic attribute knowledge)			
Evaluation metrics	T-GCN	ST-KG (Static attribute knowledge) 3D Deformation direction	Dry beach length	Reservoir water level	Phreatic line	Precipitation
RMSE	0.720	0.716	0.400	0.399	0.464	0.398
MAE	0.362	0.353	0.215	0.212	0.240	0.211
Accuracy	0.944	0.949	0.965	0.961	0.961	0.965
R^2	0.968	0.973	0.987	0.984	0.985	0.988
VAR	0.968	0.973	0.987	0.984	0.985	0.988

(b) Unembedded ST-KG vs. model embedded with different attributes (all static attributes/all dynamic attributes/all static + all dynamic attributes)

		Scene-GCN		
Evaluation metrics	T-GCN	ST-KG (Static attribute knowledge)	ST-KG (Dynamic attribute knowledge)	ST-KG (Static + Dynamic attribute knowledge)
RMSE	0.720	0.716	0.391	**0.366**
MAE	0.362	0.353	0.215	**0.196**
Accuracy	0.944	0.949	0.962	**0.965**
R^2	0.968	0.973	0.985	**0.987**
VAR	0.968	0.973	0.986	**0.987**

The experimental results show that the Scene-GCN method achieves better performance than traditional methods such as ARIMA and SVR as well as deep learning methods such as GRU, A3T-GCN, and T-GCN. The former model group cannot adapt to situations involving complex changes in deformation, while the latter group only considers time-dependency or spatial–temporal dependency and does account for the influence of geographic knowledge on deformation generation. The results indicate that the RMSE value of the Scene-GCN (Static + Dynamic) model is 97.5%, 79.8%, 61.1%, 46.4% and 49.2% lower than that of ARIMA, SVR, GRU, A3T-GCN, and T-GCN, respectively.

(2) Ablation experiments

The ablation experiments performed here confirm that the dynamic and static attributes in the Scene-GCN can improve the forecasting performance of deep learning models. The ablation experiments were conducted using the settings described above, which are categorized as no ST-KG embedded, only a single static or dynamic attribute embedded, all static or dynamic attributes embedded, and all static and dynamic attributes embedded. The results of this analysis are shown in Table 2.

As shown in Table 2(a), compared to the T-GCN without embedded ST-KG, the Scene-GCN models with a single embedded attribute exhibited great improvements in all evaluation metrics. Specifically, the RMSE decreased by 0.6% when embedding a single static attribute, while the average RMSE decreased by 42.3% when embedding a single dynamic attribute. Similarly, Table 2(b) shows that the Scene-GCN embedded with different attributes (static only/dynamic only/static + dynamic) exhibited significant improvements in all evaluation metrics compared to the T-GCN model without embedded ST-KG. Among these, the accuracy of the evaluation metrics embedded with the static attributes is as described above; the RMSE of the model embedded with dynamic attributes decreased by 45.8%, and the RMSE of the model embedded with both the static and dynamic attributes decreased by 49.2%.

To further investigate the effectiveness of the Scene-GCN approach, we randomly selected a single monitoring point from the monitoring network and visualized the results of the ablation experiments for its time-series data. The comparisons of the predicted visualizations are shown in Figures 9–11.

As shown in Figures 9–11, embedding the ST-KG improves the model's ability to perceive peaks and inflection points. In addition, when the deformation monitoring data and the model prediction data have the same monotonicity, the fit of the predicted deformation is improved and the predicted results are closer to the ground truth.

From the visualization results shown in Figure 12, relative to the ground truth, the deviations in the forecast results from the model with both dynamic and static attributes embedded are smaller than those from models with only static or only dynamic

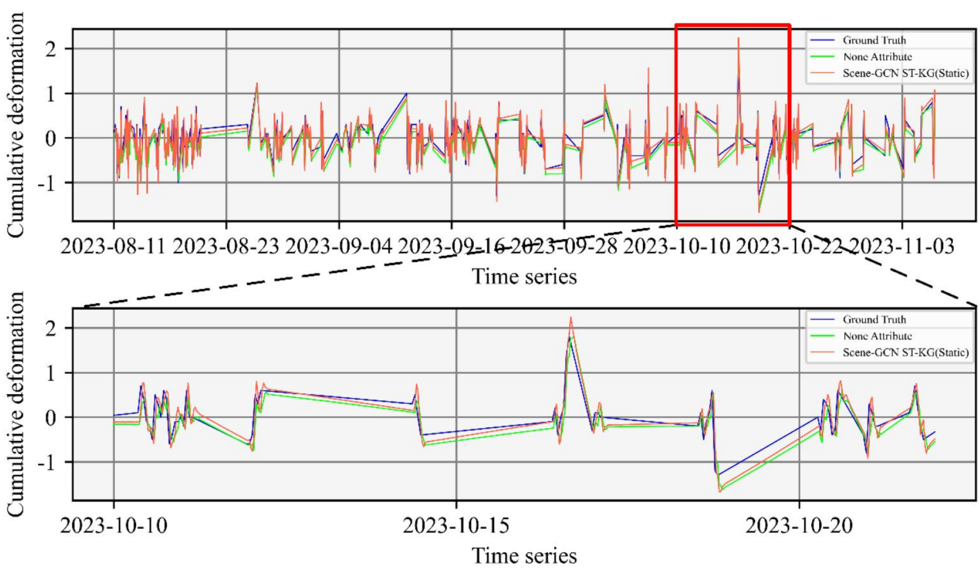

Figure 9. Comparison between model forecasting with embedded static attributes and forecasting with no attributes.

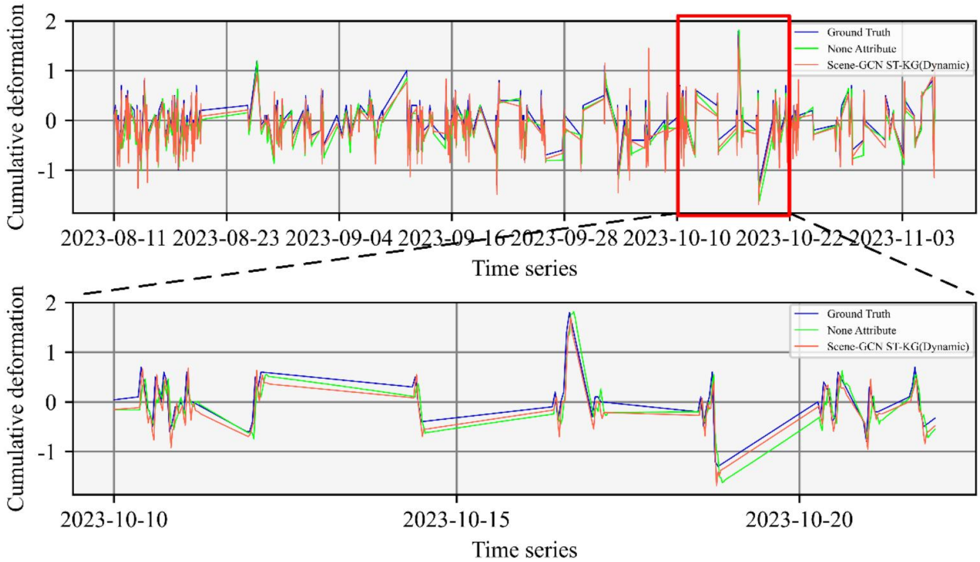

Figure 10. Comparison between model forecasting with embedded dynamic attributes and forecasting with no attributes.

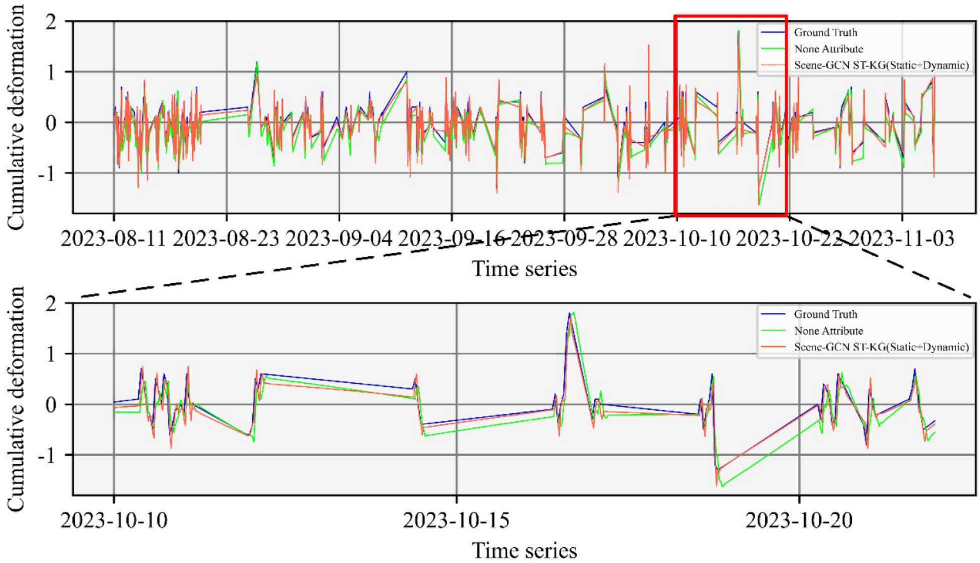

Figure 11. Comparison between model forecasting with embedded static and dynamic attributes and forecasting with no attributes.

attributes, with more accurate predictions at peaks and inflection points. This suggests that incorporating diverse geographic knowledge when constructing the ST-KG can improve the model's forecasting accuracy in complex geographic scenes.

Based on the above analysis, Scene-GCN models that consider only single or partial attributes are unable to effectively capture the implicit relationship between deformation and the influencing attributes, making it challenging to further improve forecast

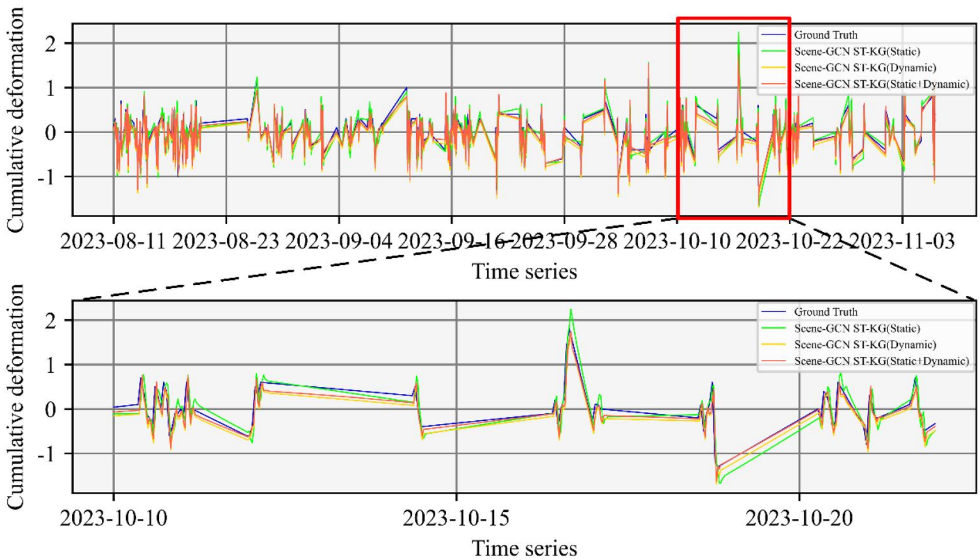

Figure 12. Comparison of embedding different state attributes for forecasting.

accuracy. In contrast, the Scene-GCN model that incorporates all attributes performs better. This indicates that the inclusion of attribute knowledge related to the prediction targets enriches the structure of the ST-KG, enabling Scene-GCN to more comprehensively capture the spatio-temporal dependencies associated with the prediction targets. Therefore, this process clearly demonstrates that the improvement in Scene-GCN's prediction capability is attributable to its ability to incorporate more attribute knowledge, thereby better explaining the changes and trends of the prediction targets. In other words, it enhances the interpretability of the deep learning model in deformation prediction. Therefore, embedding the ST-KG approach significantly improves both the accuracy and interpretability of deep learning models in complex geographic scenes.

5. Conclusion

In this study, Scene-GCN time-series prediction model based on the ST-KG approach is proposed for monitoring networks in complex geographic scenes. Our model overcomes the limitations of traditional forecasting methods by providing the monitoring network in complex geographic scenes with a spatial structure suitable for graph model learning. It combines dynamic and static environmental knowledge of the scene with the dynamic changes in time-series data from the monitoring targets, constructing a "process–state" ST-KG that allows the model to incorporate various attributes that may influence deformation. Additionally, the T-GCN cell captures the spatial–temporal dependencies within the monitoring network by learning time-series and attribute information from different monitoring points within the ST-KG. Experimental results demonstrate that the Scene-GCN outperforms baseline models across all studied prediction tasks. Furthermore, a series of ablation experiments validate the effectiveness of integrating the ST-KG within our proposed model. As the T-GCN model exhibits robust performance in both short- and long-term predictions (Zhao *et al*. 2020), thereby empowering the Scene-GCN framework to enhance real-time responsiveness while enabling risk assessment and long-term planning.

The effectiveness of the proposed method depends on two key factors: (1) the precise construction of adjacency matrices and (2) the accurate identification of environmentally influential factors. When both are well-defined, Scene-GCN demonstrates strong performance across various geographic scales. In small-scale environments, such as tailings ponds, clearly defined spatial relationships and homogeneous environmental conditions enable Scene-GCN to achieve state-of-the-art performance. However, scaling Scene-GCN to larger, more heterogeneous geographic scenes presents significant challenges. As monitoring points extend across diverse topographical features, such as rivers and valleys, determining spatial correlations becomes increasingly complex, reducing adjacency matrix accuracy. Additionally, extracting relevant environmental factors in large-scale scenes is further complicated by spatial variability, making it difficult to identify common environmental patterns and integrate accurate data into the framework. To enhance the model's applicability, future research will focus on developing tools for adjacency matrix construction and methods for rapidly selecting common environmental factors in large-scale geographic scenes.

Acknowledgement

We sincerely thank the editors and reviewers for their insightful comments and constructive suggestions, which have significantly improved the quality of this paper.

Author contributions

Weihao Li formulated the collected data, conducted the experiment, and wrote the manuscript. Yufeng He overarching research goals, contributed to the concept of the current research and provided experimental ideas. Lizhi Tao and Jifa Chen formatted the manuscript and corrected any errors observed. Hui Lin and Yong Ge was responsible for the quality audit of the manuscript.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by [National Natural Science Foundation of China] under Grants [42301533, 42230108 and 42471461], [China Postdoctoral Science Foundation] under Grant [2024M751558], [Natural Science Foundation of Jiangxi Province] under Grant [20242BAB25118], [Key Research and Development Program Project of Jiangxi Province] under Grant [20243BBI91033].

Notes on contributors

Weihao Li is currently pursuing the M.S. degree at Jiangxi Normal University. His research directions are geographic knowledge graph, graph neural networks, and spatio-temporal forecasting.

Yufeng He is currently an associate professor at Jiangxi Normal University. His research interests include geoinformatics, spatiotemporal data models, and geographic big data mining. At present, his main research directions are geographic knowledge graph and Virtual Geographic Environments/Geo-Scene.

Lizhi Tao is currently a lecturer at Jiangxi Normal University. Her research interests include spatio-temporal big data mining, time series analysis and modeling.

Jifa Chen is currently a lecturer at Jiangxi Normal University. His research interests include spatio-temporal big data mining, time series analysis and modeling.

Xing Wang is a postdoctoral researcher in Nanjing University. His research interests include Multimodal data analysis, Wireless sensor network, Deep learning, and Video-GIS.

Yong Ge is a researcher at Jiangxi Normal University, a researcher at the Institute of Geographic Sciences and Resources, Chinese Academy of Sciences (IGSR, CAS), a recipient of the National Distinguished Youth Fund (NYF), a chairperson of the Specialised Committee on Cartography and Geographic Information Systems (SCGIS) of the Geographical Society of China (GSC), and the secretary-general of the Specialised Committee on Data-Driven and Geological Development (SDGD) of the Chinese Geological Society (CGS). She is currently employed as Associate Editor of Computers & Geosciences, Associate Editor of Journal of Remote Sensing, and Editorial Board Member of Spatial Statistics, among others. She has been honored as the Most Beautiful Geoscientist and one of the top 2% scientists in the world at Stanford University.

Hui Lin is Professor and Dean of the School of Geography and Environment, Jiangxi Normal University. He is mainly engaged in Virtual Geographic Environments, Remote Sensing in Tropical and Sub-Tropical Areas, Spatially Integrated Humanities and Social Science.

Data and codes availability statement

The dataset used in this study consists of monitoring network data from August 25, 2022, to November 6, 2023, for the Dexing Copper Mine No. 4 tailings pond. The model was constructed by using Python language in Jupyter notebook. The study was conducted using Tensorflow version 1.15, operating system ubuntu 20.04, 32 G of RAM and 1 T hard disk.

The model and data code of the paper is provided in the following link: https://github.com/Jolisen/Scene-GCN

References

Bai, J., *et al.*, 2021. A3t-gcn: Attention temporal graph convolutional network for traffic forecasting. *ISPRS International Journal of Geo-Information*, 10 (7), 485.

Bui., *et al*., 2019. Predicting slope stability failure through machine learning paradigms. *ISPRS International Journal of Geo-Information*, 8 (9), 395.

Cho, K., *et al.*, 2014. On the properties of neural machine translation: encoder-decoder approaches. *arXiv preprint arXiv:1409.1259*.

Chung, J., *et al.*, 2014. Empirical evaluation of gated recurrent neural networks on sequence modeling. *arXiv preprint arXiv:1412.3555*.

da Cruz, M.A., *et al.*, 2018. A reference model for internet of things middleware. *IEEE Internet of Things Journal*, 5 (2), 871–883.

Defferrard, M., Bresson, X., and Vandergheynst, P., 2016. Convolutional neural networks on graphs with fast localized spectral filtering. *Advances in Neural Information Processing Systems*, 29.

Diskin, M., 1970. Definition and uses of the linear regression model. *Water Resources Research*, 6 (6), 1668–1673.

Dong, L., Deng, S., and Wang, F., 2020. Some developments and new insights for environmental sustainability and disaster control of tailings dam. *Journal of Cleaner Production*, 269, 122270.
Du, S. G., *et al.*, 2022. A new approach for evaluation of slope stability in large open-pit mines: a case study at the Dexing Copper Mine, China. *Environmental Earth Sciences*, 81 (3), 102.
Elsheikh, A., Yacout, S., and Ouali, M.S., 2019. Bidirectional handshaking LSTM for remaining useful life prediction. *Neurocomputing*, 323, 148–156.
Elshorbagy, A., *et al.*, 2010. Experimental investigation of the predictive capabilities of data driven modeling techniques in hydrology - Part 2: Application. *Hydrology and Earth System Sciences*, 14 (10), 1943–1961.
Gama, F., *et al.*, 2020. Deformations prior to the Brumadinho Dam collapse revealed by sentinel-1 InSAR data using SBAS and PSI techniques. *Remote Sensing*, 12 (21), 3664.
Gao, J., and Fourie, A., 2015. Using the flume test for yield stress measurement of thickened tailings. *Minerals Engineering*, 81, 116–127.
Gao, W., *et al.*, 2022. Modelling and prediction of GNSS time series using GBDT, LSTM and SVM machine learning approaches. *Journal of Geodesy*, 96 (10), 71.
Hancock, G.R., 2021. A method for assessing the long-term integrity of tailings dams. *The Science of the Total Environment*, 779, 146083.
He, Y., *et al.*, 2023. Processes and events in the center: a taxi trajectory-based approach to detecting traffic congestion and analyzing its causes. *International Journal of Digital Earth*, 16 (1), 509–531.
He, Y., *et al.*, 2025. SceneKG: a geo-scene based spatiotemporal knowledge representation framework considering geo-processes. *Transactions in GIS*, 29 (1), e13270.
He, Y., *et al.*, 2022. Processes and events in the centre: a dynamic data model for representing spatial change. *International Journal of Digital Earth*, 15 (1), 276–295.
Hua, G., *et al.*, 2023. Research on the uplift pressure prediction of concrete dams based on the CNN-GRU model. *Water*, 15 (2), 319.
Huang, Y., *et al.*, 2019. Using geographic ontologies and geo-characterization to represent geographic scenarios. *ISPRS International Journal of Geo-Information*, 8 (12), 566.
Ji, P., *et al.*, 2022. A new method to obtain 3-D surface deformations from InSAR and GNSS data with genetic algorithm and support vector machine. *IEEE Geoscience and Remote Sensing Letters*, 19, 1–5.
Jiang, Y., *et al.*, 2024. Deformation mechanism-assisted deep learning architecture for predicting step-like displacement of reservoir landslide. *International Journal of Applied Earth Observation and Geoinformation*, 133, 104121.
Kingma, D.P., and Ba, J., 2014. Adam: a method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.
Kipf, T.N., and Welling, M., 2016. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*.
Kochkov, D., *et al.*, 2024. Neural general circulation models for weather and climate. *Nature*, 632 (8027), 1060–1066.
Krahn, J., 2003. The 2001 RM Hardy lecture: the limits of limit equilibrium analyses. *Canadian Geotechnical Journal*, 40 (3), 643–660.
Li, J., *et al.*, 2019. Tailings pond risk prediction using long short-term memory networks. *IEEE Access*, 7, 182527–182537.
Li, Y., 2011. Web-based GNSS deformation monitoring software for landslide and dam. ed. *2011 International Conference on Remote Sensing, Environment and Transportation Engineering*, 2825–2827.
Lin, X., *et al.*, 2023. A hybrid deep learning model for rapid probabilistic earthquake source parameter estimation with displacement waveforms from a flexible set of seismic or HR-GNSS stations. *IEEE Transactions on Geoscience and Remote Sensing*, 61, 1–16.
Lü, G., *et al.*, 2018. Geographic scenario: a possible foundation for further development of virtual geographic environments. *International Journal of Digital Earth*, 11 (4), 356–368.

Ma, Z., *et al.*, 2021. A deep learning approach using graph convolutional networks for slope deformation prediction based on time-series displacement data. *Neural Computing and Applications*, 33 (21), 14441–14457.

Mendoza, R., *et al.*, 2021. Estimation of the Pb content in a tailings dam using a linear regression model based on the chargeability and resistivity values of the wastes (La Carolina Mining District, Spain). *Minerals*, 12 (1), 7.

Metternicht, G., Hurni, L., and Gogu, R., 2005. Remote sensing of landslides: An analysis of the potential contribution to geo-spatial systems for hazard assessment in mountainous environments. *Remote Sensing of Environment*, 98 (2-3), 284–303.

Moravčík, M., *et al.*, 2017. Deepstack: Expert-level artificial intelligence in heads-up no-limit poker. *Science (New York, N.Y.)*, 356 (6337), 508–513.

Naeini, M., and Akhtarpour, A., 2018. Numerical analysis of seismic stability of a high centerline tailings dam. *Soil Dynamics and Earthquake Engineering*, 107, 179–194.

Nie, W., *et al.*, 2022. 3D visualization monitoring and early warning system of a tailings dam—gold copper mine tailings dam in Zijinshan, Fujian, China. *Frontiers in Earth Science*, 10, 800924.

Percival, D.B., and Walden, A.T., 1993. *Spectral analysis for physical applications*. Cambridge: Cambridge University Press.

Qin, L., Feng, S., and Zhu, H., 2018. Research on the technological architectural design of geological hazard monitoring and rescue-after-disaster system based on cloud computing and Internet of things. *International Journal of System Assurance Engineering and Management*, 9 (3), 684–695.

Rose, N.D., and Hungr, O., 2007. Forecasting potential rock slope failure in open pit mines using the inverse-velocity method. *International Journal of Rock Mechanics and Mining Sciences*, 44 (2), 308–320.

Ruan, S., *et al.*, 2023. Proactive control model for safety prediction in tailing dam management: Applying graph depth learning optimization. *Process Safety and Environmental Protection*, 172, 329–340.

Salazar, F., *et al.*, 2015. An empirical comparison of machine learning techniques for dam behaviour modelling. *Structural Safety*, 56, 9–17.

Shen, N., Chen, L., and Chen, R., 2022. Displacement detection based on Bayesian inference from GNSS kinematic positioning for deformation monitoring. *Mechanical Systems and Signal Processing*, 167, 108570.

Silva Rotta, L.H., *et al.*, 2020. The 2019 Brumadinho tailings dam collapse: Possible cause and impacts of the worst human and environmental disaster in Brazil. *International Journal of Applied Earth Observation and Geoinformation*, 90, 102119.

Silver, D., *et al.*, 2016. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529 (7587), 484–489.

Smola, A.J., and Schölkopf, B., 2004. A tutorial on support vector regression. *Statistics and Computing*, 14 (3), 199–222.

Sun, E., Zhang, X., and Li, Z., 2012. The internet of things (IOT) and cloud computing (CC) based tailings dam monitoring and pre-alarm system in mines. *Safety Science*, 50 (4), 811–815.

Van Lint, J., and Van Hinsbergen, C., 2012. Short-term traffic and travel time prediction models. *Artificial Intelligence Applications to Critical Transportation Issues*, 22 (1), 22–41.

Vlahogianni, E.I., Golias, J.C., and Karlaftis, M.G., 2004. Short-term traffic forecasting: overview of objectives and methods. *Transport Reviews*, 24 (5), 533–557.

Von Luxburg, U., 2007. A tutorial on spectral clustering. *Statistics and Computing*, 17 (4), 395–416.

Wang, J., Li, G., and Ai, T., 2024. Revealing association rules within intricate ecosystems: A spatial co-location mining method based on Geo-Eco knowledge graph. *International Journal of Applied Earth Observation and Geoinformation*, 133, 104116.

Wu, Z., *et al.*, 2021. A comprehensive survey on graph neural networks. *Ieee Transactions on Neural Networks and Learning Systems*, 32 (1), 4–24.

Yang, B., *et al.*, 2023. Updated global navigation satellite system observations and attention-based convolutional neural network–long short-term memory network deep learning algorithms to predict landslide spatiotemporal displacement. *Remote Sensing*, 15 (20), 4971.

Yang, J., *et al.*, 2020. A CNN-LSTM model for tailings dam risk prediction. *IEEE Access*, 8, 206491–206502.

Yang, J., *et al.*, 2019. Measure dry beach length of tailings pond using deep learning algorithm. ed. *Proceedings of the 2019 International Conference on Robotics, Intelligent Control and Artificial Intelligence*, 503–508.

Yao, L., Mao, C., and Luo, Y., 2019. Graph convolutional networks for text classification. ed. *Proceedings of the AAAI conference on artificial intelligence*, 7370–7377.

Yin, G., *et al.*, 2011. Stability analysis of a copper tailings dam via laboratory model tests: A Chinese case study. *Minerals Engineering*, 24 (2), 122–130.

Zhang, M., *et al.*, 2021. Continuous detection of surface-mining footprint in copper mine using google earth engine. *Remote Sensing*, 13 (21), 4273.

Zhang, W., *et al.*, 2022. Displacement prediction of Jiuxianping landslide using gated recurrent unit (GRU) networks. *Acta Geotechnica*, 17 (4), 1367–1382.

Zhao, L., *et al.*, 2020. T-GCN: a temporal graph convolutional network for traffic prediction. *IEEE Transactions on Intelligent Transportation Systems*, 21 (9), 3848–3858.

Zhou, J., *et al.*, 2020. Graph neural networks: a review of methods and applications. *AI Open*, 1, 57–81.

Zhu, Y., *et al.*, 2022. A tailings dam long-term deformation prediction method based on empirical mode decomposition and LSTM model combined with attention mechanism. *Water*, 14 (8), 1229.